

LAB EXPERIMENT RECORD

Set 17: Employee Performance Analysis

Department	B.Tech AIDS
Course Code / Course Name	DSA06 / Data Handling and Visualization
Experiment	Data Visualization using R
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Aim

To analyze employee performance data using a line chart, bar chart, and table in R, and to build an interactive R Shiny dashboard combining all three.

Dataset: Employee Performance

Employee ID	Department	Years of Service	Performance Score
1	Sales	5	85
2	HR	3	92
3	Marketing	7	78

Note: The dataset has no date column, so the employees are taken in order of Employee ID to represent the trend over time in Question 1.

PROGRAM 1: Line Chart of Performance Trend

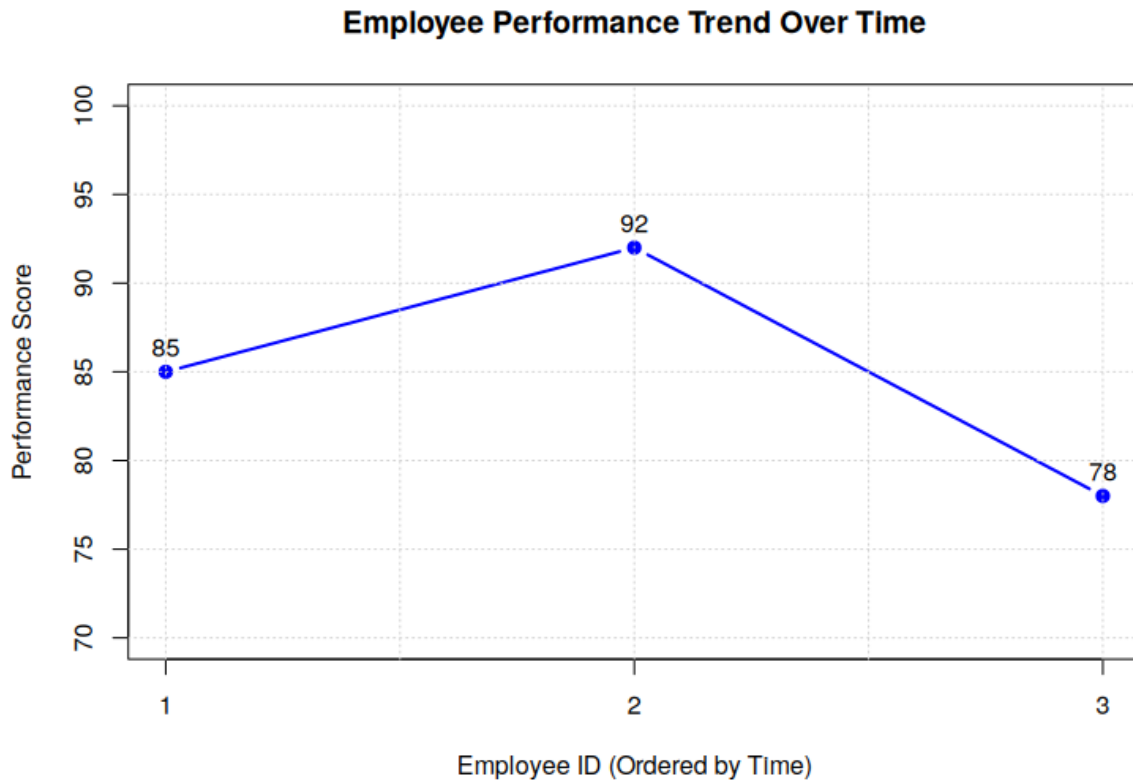
Create a line chart to visualize the performance trend of employees over time. Label the axes and title the chart.

```
# Create dataset for Set 17
employee_data <- data.frame(
  Employee_ID = c(1, 2, 3),
  Department = c("Sales", "HR", "Marketing"),
  Years_of_Service = c(5, 3, 7),
  Performance_Score = c(85, 92, 78)
)

print(employee_data)

# 1. Line Chart of Performance Trend
plot(employee_data$Employee_ID, employee_data$Performance_Score,
      type = "b",
      col = "blue",
      lwd = 2,
      pch = 19,
```

```
xaxt = "n",
ylim = c(70, 100),
xlab = "Employee ID (Ordered by Time)",
ylab = "Performance Score",
main = "Employee Performance Trend Over Time")
axis(1, at = employee_data$Employee_ID)
text(employee_data$Employee_ID, employee_data$Performance_Score,
      labels = employee_data$Performance_Score, pos = 3)
grid()
```



Output 1: Employee Performance Trend Over Time

Observation: The performance score rises from 85 (Employee 1) to the highest value of 92 (Employee 2) and then falls to the lowest value of 78 (Employee 3). The average performance score is 85.

PROGRAM 2: Bar Chart of Employees by Department

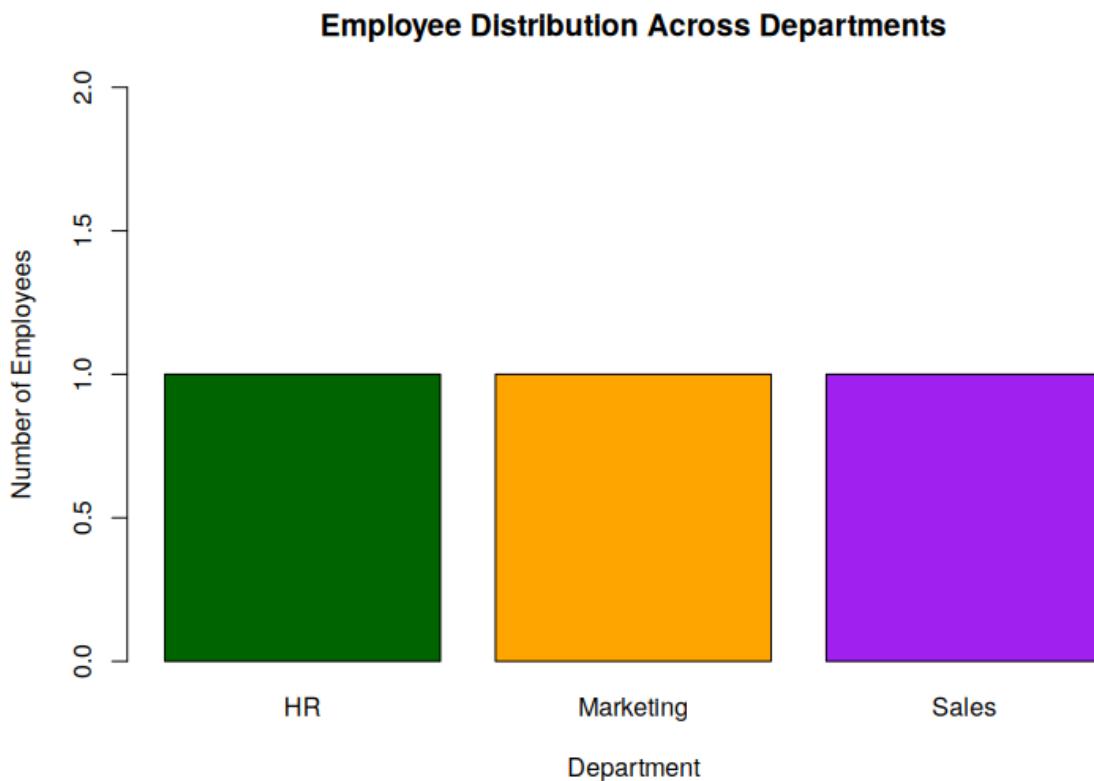
Generate a bar chart showing the distribution of employees across different departments. Label the chart elements.

```
# Create dataset for Set 17
employee_data <- data.frame(
  Employee_ID = c(1, 2, 3),
  Department = c("Sales", "HR", "Marketing"),
  Years_of_Service = c(5, 3, 7),
  Performance_Score = c(85, 92, 78)
)

print(employee_data)

# Count employees per department
dept_counts <- table(employee_data$Department)

# 2. Bar Chart of Employees by Department
barplot(dept_counts,
  col = c("darkgreen", "orange", "purple"),
  xlab = "Department",
  ylab = "Number of Employees",
  main = "Employee Distribution Across Departments",
  ylim = c(0, 2))
```



Output 2: Employee Distribution Across Departments

Observation: Each department (HR, Marketing, and Sales) has exactly one employee, so the employees are evenly distributed across the three departments.

PROGRAM 3: Table of Employee Performance Data

Build a table to display the performance data for each employee. Label the table elements.

```
# Create dataset for Set 17
employee_data <- data.frame(
  Employee_ID = c(1, 2, 3),
  Department = c("Sales", "HR", "Marketing"),
  Years_of_Service = c(5, 3, 7),
  Performance_Score = c(85, 92, 78)
)

print(employee_data)

# 3. Table of employee performance data
perf_table <- employee_data
colnames(perf_table) <- c("Employee ID", "Department",
                          "Years of Service", "Performance Score")

cat("=== EMPLOYEE PERFORMANCE TABLE ===\n")
print(perf_table, row.names = FALSE)
```

```
=== EMPLOYEE PERFORMANCE TABLE ===
> print(perf_table, row.names = FALSE)
  Employee ID Department Years of Service Performance Score
          1      Sales              5             85
          2         HR              3             92
          3  Marketing              7             78
>
```

Output 3: Employee Performance Table

Observation: The table shows the department, years of service, and performance score of each employee with clearly labelled columns. The HR employee has the highest score (92) with only 3 years of service, while the Marketing employee has the longest service (7 years) but the lowest score (78).

PROGRAM 4: Interactive Dashboard for Employee Performance

Develop a Tableau dashboard combining the line chart, bar chart, and the table for interactive exploration of employee performance data.

Note: Tableau could not be installed on the system used for this experiment, so the interactive dashboard was developed in R using the Shiny package, with the same charts and interactive filters required in the question.

```
library(shiny)

# Dataset for Set 17
employee_data <- data.frame(
  Employee_ID = c(1, 2, 3),
  Department = c("Sales", "HR", "Marketing"),
  Years_of_Service = c(5, 3, 7),
  Performance_Score = c(85, 92, 78)
)

ui <- fluidPage(
  titlePanel("Employee Performance Dashboard"),
  sidebarLayout(
    sidebarPanel(width = 3,
      checkboxGroupInput("dept", "Department:",
        choices = c("HR", "Marketing", "Sales"),
        selected = c("HR", "Marketing", "Sales"))
    ),
    mainPanel(width = 9,
      fluidRow(
        column(6, plotOutput("linePlot", height = "300px")),
        column(6, plotOutput("barPlot", height = "300px"))
      ),
      h4("Employee Performance Table"),
      tableOutput("perfTable")
    )
  )
)

server <- function(input, output) {
  filtered <- reactive({ subset(employee_data, Department %in% input$dept) })

  output$linePlot <- renderPlot({
    d <- filtered()
    plot(d$Employee_ID, d$Performance_Score, type = "b", col = "blue", lwd = 2, pch = 19,
      xlim = c(1, 3), ylim = c(70, 100), xaxt = "n",
      xlab = "Employee ID (Ordered by Time)", ylab = "Performance Score",
      main = "Employee Performance Trend")
    axis(1, at = 1:3)
    grid()
  }, width = 420, height = 300, res = 96)

  output$barPlot <- renderPlot({
    counts <- table(factor(filtered()$Department, levels = c("HR", "Marketing", "Sales")))
    barplot(counts, col = c("darkgreen", "orange", "purple"), ylim = c(0, 2),
      xlab = "Department", ylab = "Number of Employees",
      main = "Employees by Department")
  }, width = 420, height = 300, res = 96)
```

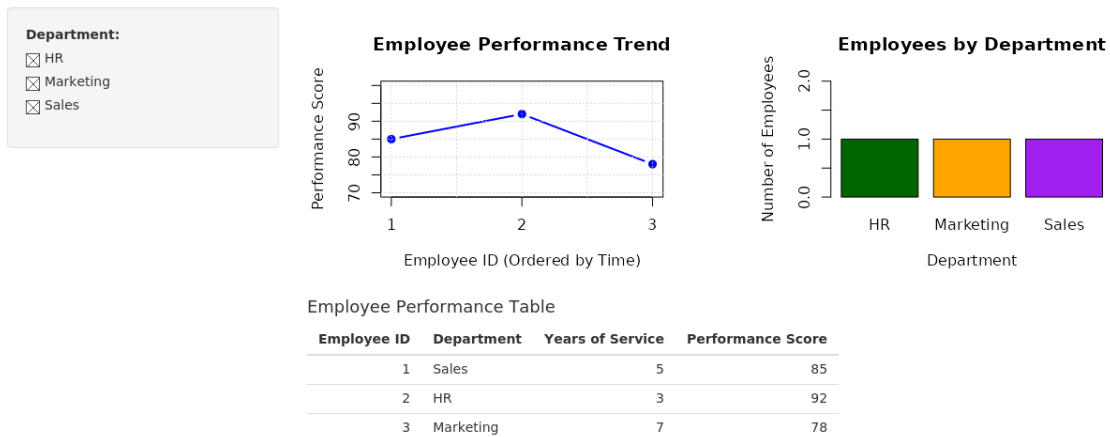
```

output$perfTable <- renderTable({
  d <- filtered()
  colnames(d) <- c("Employee ID", "Department", "Years of Service", "Performance Score")
  d
}, digits = 0)
}

shinyApp(ui = ui, server = server)

```

Employee Performance Dashboard



Output 4: Employee Performance Interactive Dashboard (R Shiny)

Observation: The dashboard shows the performance trend, the department bar chart, and the performance table together. Unticking a department in the Department filter updates all three views at once.

Procedure

1. Create the employee data frame with Employee ID, Department, Years of Service, and Performance Score.
2. Use `plot()` with `type = "b"` to draw the performance line chart and `text()` to label each point.
3. Count employees per department with `table()` and use `barplot()` for the bar chart.
4. Rename the columns with `colnames()` and print the labelled performance table.
5. Install and load the shiny package, and create the dashboard layout using `fluidPage()`, sidebar filters, and output areas for the charts and table.
6. Write the server function to filter the data with `reactive()` and redraw the outputs, then run the dashboard using `shinyApp()` and test the filters.

Result

The employee performance data was successfully visualized using a line chart, bar chart, and table in R, and an interactive R Shiny dashboard was built combining all three views.

Key R Functions Used

Function	Purpose
<code>data.frame()</code>	Creates the employee dataset.
<code>plot()</code>	Creates the performance line chart.
<code>text()</code>	Adds score labels to the points.
<code>table()</code>	Counts employees in each department.
<code>barplot()</code>	Creates the department bar chart.
<code>colnames()</code> / <code>print()</code>	Labels and displays the performance table.
<code>fluidPage()</code> / <code>checkboxGroupInput()</code>	Creates the dashboard layout and filters.
<code>reactive()</code> / <code>renderPlot()</code>	Filters the data and redraws the charts.
<code>renderTable()</code>	Displays the filtered table on the dashboard.
<code>shinyApp()</code>	Runs the interactive dashboard.